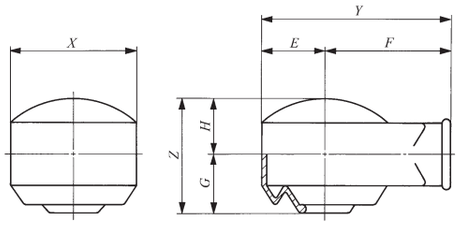
L-BALL

L-Ball Dust Cover





PRC

Identification number	Boundary dimensions mm						
	X	*Y*	*E*	*F*	*Z*	*G*	*H*
PRC 5	20	29	10	19	16	8	8
PRC 6	22	31	11	20	19	9.5	9.5
PRC 8	27	38.5	13.5	25	24	12	12
PRC 10	31	45.5	15.5	30	27	14	13
PRC 12	36	53	18	35	32	16.5	15.5
PRC 14	40	60	20	40	36.5	19	17.5
PRC 16	44	68	22	46	40	20.5	19.5
PRC 18	49	74.5	24.5	50	46	23.5	22.5
PRC 20	54	82	27	55	50	25.5	24.5
PRC 22	59	89.5	29.5	60	53.5	27.5	26



1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

SUPER FLEXIBLE NOZZLES



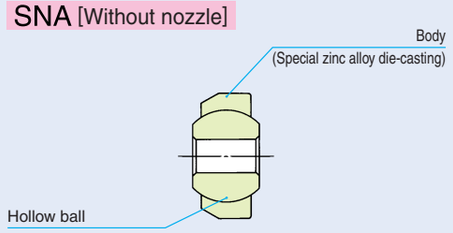
Structure and Features

IKO Super Flexible Nozzle is a compact nozzle for use on a machine tool to supply and spray cutting oil exactly at the required positions.
The angle of the nozzle can be changed easily and freely. Therefore, oil supply can be concentrated upon the working area, and cooling and lubrication can be performed effectively. As a result, cutting resistance is reduced and superior finish is obtained, achieving high machining accuracy. Also, tool life is longer.
The Super Flexible Nozzle is used in many places such as at the spindle end of Machining Center and at the tool holder of N/C lathe.

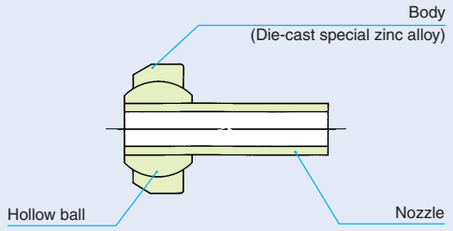
The features of Super Flexible Nozzle are as follows.
❶A spherical bushing is incorporated to adjust the tilting angle of nozzle easily.
❷The Super Flexible Nozzle is compact in size, and the design on parts around the spindle and tool can be made simple.
❸The nozzle length is short, and winding of cutting chips around the nozzle will not occur.
❹By using a number of Super Flexible Nozzles, cutting oil can be supplied and cutting chips can be removed more effectively.
❺The press fitting type and screw fitting type are available. The press fitting type is economical.

Structures of Super Flexible Nozzles

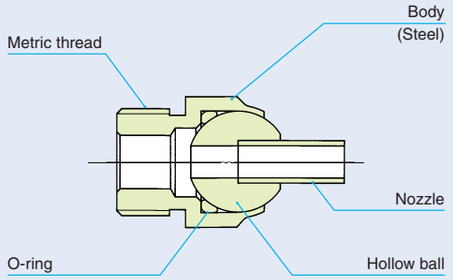
SNA [Without nozzle]



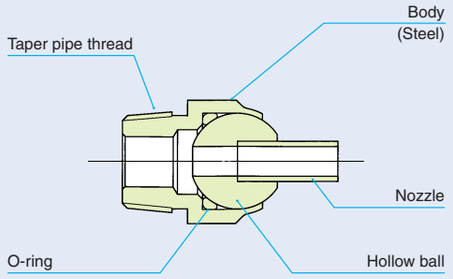
SNA [With nozzle]



SNM



SNPT



Types

Super Flexible Nozzles shown in Table 1 are available.

Table 1 Type of Super Flexible Nozzle

Type		Model code
Press fitting type	Without nozzle	SNA
	With nozzle	
Screw fitting type	With metric threads	SNM
	With taper pipe threads	SNPT

Identification Number

The identification number of Super Flexible Nozzle consists of a model code and a size. An example is shown as follows.

Example of identification number

Model code	Size
SNM	10−20

- Type of nozzle
- Nozzle bore or thread size: (M10×1.25)
- Dimension from shoulder surface to nozzle top:※ (20mm)

※In case of press fitting type without nozzle, this dimension is not indicated.

Precautions for Use

When the press fitting type Super Flexible Nozzle is used, a ϕ 15 (H8) $^{+0.027}_{0}$ bore for fitting hole must be prepared and fitting is made from the 30° chamfered end of the outer body. In this case, the body portion should be pushed for press fitting.

When the screw fitting type Super Flexible Nozzle is used and prevention of oil leakage from the fitting part is required, it is recommended to wind sealing tape on the thread portion or use rubber packing for the shoulder face of the outer body.

The direction of lubrication can be adjusted by inserting a screwdriver, etc. in the bore of the nozzle.

Special Specifications

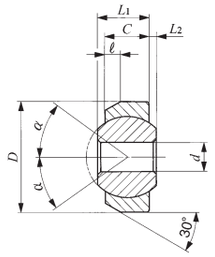
Super Flexible Nozzles with special length are also available. In this case, specify the necessary nozzle length in units of 1 mm, but do not exceed the maximum length shown in the dimension table as "L".

Super Flexible Nozzles with curved nozzle end or with special bore diameter are also available. In this case, please contact IKO by preparing a drawing or sketch with necessary specifications.

SUPER FLEXIBLE NOZZLE

Press Fitting Type Without Nozzle

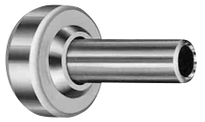


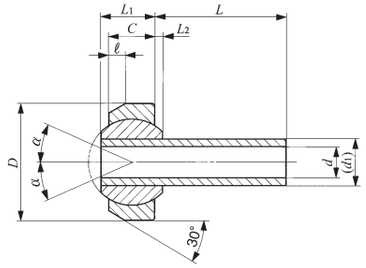


SNA

Identification number	Boundary dimensions mm d	D	L_1	L_2	C	ℓ	Ball dia. mm (inch)	Allowable tilting angle α degree
SNA 4	4	15	7	1	6	2	11.112 (7/16)	36
SNA 6	6							24

Press Fitting Type With Nozzle





SNA

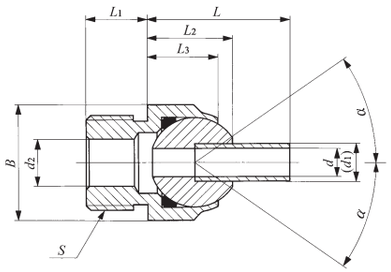
Identification number	Boundary dimensions mm d	D	L	L_1	L_2	C	ℓ	d_1	Ball dia. mm (inch)	Allowable tilting angle α degree
SNA 3-*L*	3	15	6 15 32	7	1	6	2	6	11.112 (7/16)	24
SNA 4-*L*	4		6 16 40							

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

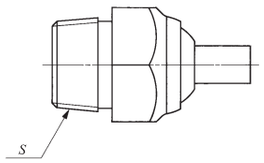
SUPER FLEXIBLE NOZZLE

Screw Fitting Type





SNM



SNPT

Identification number	Boundary dimensions mm										Ball dia.	Allowable tilting angle
	d	Thread S	L	L_1	L_2	L_3	d_1	d_2	Width across flats B	Width across corners (Ref.)	mm (inch)	α degree
SNM 10-*L*	4	M10 × 1.25	20 40 60	9	13	10.5	6	6	17	19.6	12.700 (1/2)	35
SNPT 1/4-*L*		PT 1/4										
SNM 20-*L*	6	M20 × 1.5	30 50 70	13	18	15	8	10	24	27.7	19.050 (3/4)	
SNPT 3/8-*L*		PT 3/8										
SNM 24-*L*	8	M24 × 2.0	40 60 80	18	23	19	10	12	32	37	25.400 (1)	
SNPT 1/2-*L*		PT 1/2										

PARTS FOR NEEDLE ROLLER BEARINGS

- **Seals for Needle Roller Bearings**
- **Cir-clips for Needle Roller Bearings**
- **Needle Rollers**



1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

Seals for Needle Roller Bearings

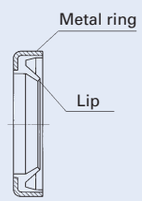
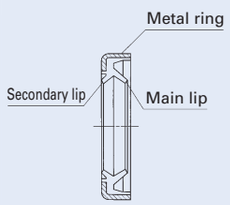
Features

IKO Seals for Needle Roller Bearings have a low sectional height and consist of a sheet metal ring and special synthetic rubber.
As these seals are manufactured to the same sectional height as IKO Needle Roller Bearings, grease leakage and the penetration of foreign particles can be effectively prevented by fitting them directly to the sides of combinable bearings shown in the dimension table.
When fitting seals to needle roller bearings with inner ring, wide inner rings (see page H2) must be used, as shown in the mounting examples.

Types

Seals for Needle Roller Bearings are available as shown in Table1.

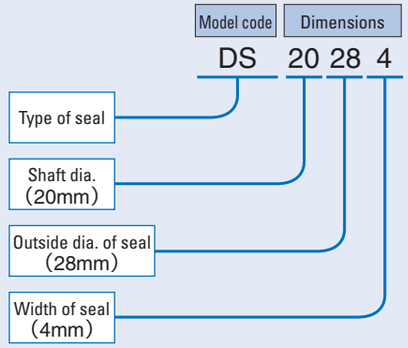
Table 1 Seal type

Type	Single lip	Double lips
Structure		
Model code	OS	DS

Identification Number

The identification number of Seals for Needle Roller Bearings consists of a model code and dimensions. An example of an identification number is shown as follows.

Example of identification number



Accuracy

Tolerances of Seals for Needle Roller Bearings are based on JIS B 2402:1996.
Tolerances of outside diameter and width are based on Tables 2 and 3, respectively.

Table 2 Tolerance of outside diameter unit: mm

Nominal outside diameter		Tolerance	
Over	Incl.	High	Low
—	30	+0.09	+0.04
30	50	+0.11	+0.05
50	80	+0.14	+0.06
80	120	+0.17	+0.08

Table 3 Tolerance of width unit: mm

Nominal size of width		Tolerance	
Over	Incl.	High	Low
—	6	+0.2	−0.2
6	10	+0.3	−0.3

Precautions for Use

❶For the single lip OS type, the lip has to face inward when using the seal to prevent grease leakage, and outward to prevent the penetration of foreign particles. The DS type of double-lips is effective for prevention of grease leakage and dust penetration. However, when the main purpose is to prevent grease leakage, the main lip should face inward, and when used mainly to prevent dust penetration, it should face outward.
❷The permissible temperature range is -20～+120℃.
For use at higher or lower temperatures, a special seal is required. Please contact IKO for further information.
❸The limiting peripheral speed of shaft depends on the conditions of use, but is normally 6 to 8 m/s.
Double this speed is possible if the conditions (lubrication, temperature, shaft finish, etc.) are good.

Mounting

When inserting the shaft, damage to the lip should be prevented by chamfering the end of the shaft, as shown in the upper part of Fig. 1. When this cannot be performed, a mounting bushing should be used, as shown in the lower part of Fig.1.
When press fitting the seal to the housing, do not strike it directly, but fit it gently, using a suitable tool.
To prevent early wear and heat generation at the seal surface, it is necessary to thickly coat the tip of the lip for the OS type, or to fill the space between the two lips for the DS type, with bearing grease.

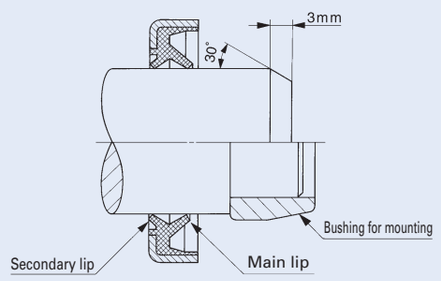


Fig.1

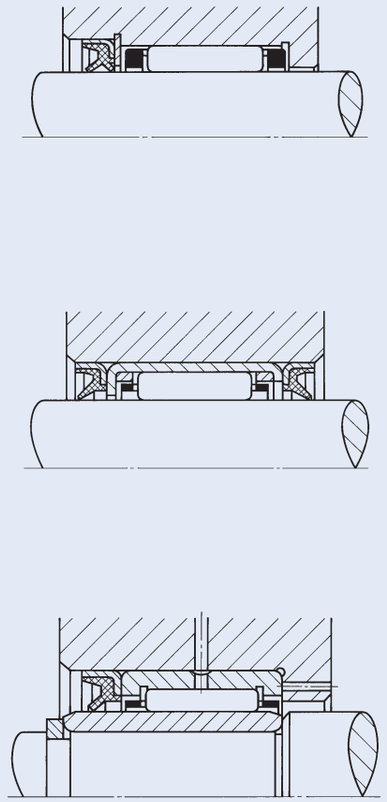


Fig.2 Mounting examples

L
OS
DS

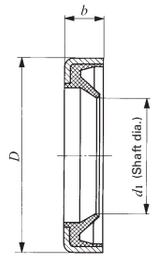
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

SEALS FOR NEEDLE ROLLER BEARINGS

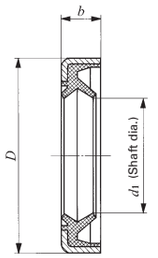


Shaft dia. 6 — 15mm

Shaft dia. mm	Identification number: Single lip	Identification number: Double lips	Boundary dimensions mm: d_1	D	b	Combinable bearings: TA···Z YT TLA···Z YTL	Combinable bearings: RNA TR TAF GTR	Combinable bearings: RNAF	Combinable bearings: NAX NBX
6	OS 6102.5	—	6	10	2.5	TLA 69Z	—	—	—
7	OS 7112.5	—	7	11	2.5	TLA 79Z	—	—	—
8	OS 8123	—	8	12	3	TLA 810Z	—	—	—
	OS 8153	—	8	15	3	TA 810Z TA 815Z TA 820Z YT 810	RNA 496 TAF 81512 TAF 81516	RNAF 81510	—
9	OS 9133	—	9	13	3	TLA 910Z TLA 912Z	—	—	—
	OS 9163	—	9	16	3	TA 912Z TA 916Z YT 912	TAF 91612 TAF 91616	—	—
10	OS 10143	—	10	14	3	TLA 1010Z TLA 1012Z TLA 1015Z	—	—	—
	OS 10173	—	10	17	3	TA 1010Z TA 1012Z TA 1015Z TA 1020Z	TAF 101712 TAF 101716	RNAF 101710	—



OS



DS

Shaft dia. mm	Identification number: Single lip	Identification number: Double lips	Boundary dimensions mm: d_1	D	b	Combinable bearings: TA···Z YT TLA···Z YTL	Combinable bearings: RNA TR TAF GTR	Combinable bearings: RNAF	Combinable bearings: NAX NBX
12	OS 12163	—	12	16	3	TLA 1210Z YTL 1210	—	—	—
	OS 12183	—	12	18	3	TLA 1212Z	—	—	—
	OS 12193	—	12	19	3	TA 1212Z TA 1215Z TA 1220Z TA 1225Z YT 1212	TAF 121912 TAF 121916	—	—
13	OS 13193	—	13	19	3	TLA 1312Z	—	—	—
14	OS 14203	DS 14203	14	20	3	TLA 1412Z TLA 1416Z	—	—	—
	OS 14223	DS 14223	14	22	3	TA 1416Z TA 1420Z	RNA 4900 TAF 142216 TAF 142220	RNAF 142213 RNAFW 142220	—
15	OS 15213	DS 15213	15	21	3	TLA 1512Z TLA 1516Z TLA 1522Z	—	—	—
	OS 15223	DS 15223	15	22	3	TA 1510Z TA 1512Z TA 1515Z TA 1520Z TA 1525Z	—	—	—
	OS 15235	DS 15235	15	23	5	—	TAF 152316 TAF 152320	RNAF 152313 RNAFW 152320	—

L

OS
DS

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

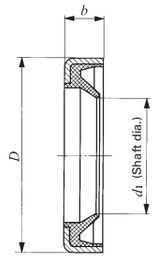
SEALS FOR NEEDLE ROLLER BEARINGS



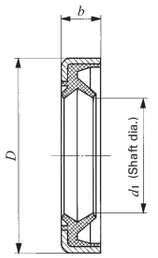


Shaft dia. 16 — 19mm

Shaft dia. mm	Identification number		Boundary dimensions mm			Combinable bearings			
	Single lip	Double lips	d_1	D	b	TA···Z YT TLA···Z YTL	RNA TR TAF GTR	RNAF	NAX NBX
16	**OS 16223**	**DS 16223**	16	22	3	**TLA 1612Z TLA 1616Z TLA 1622Z**	—	—	—
	OS 16243	**DS 16243**	16	24	3	**TA 1616Z TA 1620Z**	**RNA 4901 RNA 6901 TAF 162416 TAF 162420**	**RNAF 162413 RNAFW 162420**	—
	OS 16285	**DS 16285**	16	28	5	—	—	**RNAF 162812**	—
17	**OS 17233**	**DS 17233**	17	23	3	**TLA 1712Z**	—	—	—
	OS 17243	**DS 17243**	17	24	3	**TA 1715Z TA 1720Z TA 1725Z YT 1715 YT 1725**	—	—	—
	OS 17253	**DS 17253**	17	25	3	—	**TAF 172516 TAF 172520**	**RNAF 172513 RNAFW 172520**	—



OS



DS

Shaft dia. mm	Identification number		Boundary dimensions mm			Combinable bearings			
	Single lip	Double lips	d_1	D	b	TA···Z YT TLA···Z YTL	RNA TR TAF GTR	RNAF	NAX NBX
18	**OS 18243**	**DS 18243**	18	24	3	**TLA 1812Z TLA 1816Z**	—	—	—
	OS 18253	**DS 18253**	18	25	3	**TA 1813Z TA 1815Z TA 1817Z TA 1819Z TA 1820Z TA 1825Z**	—	—	—
	OS 18264	**DS 18264**	18	26	4	—	**RNA 49/14 TAF 182616 TAF 182620**	**RNAF 182613 RNAFW 182620**	—
19	**OS 19274**	**DS 19274**	19	27	4	**TA 1916Z TA 1920Z**	**TAF 192716 TAF 192720**	—	—

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

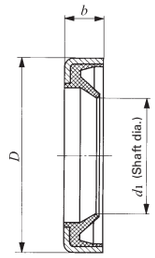
SEALS FOR NEEDLE ROLLER BEARINGS



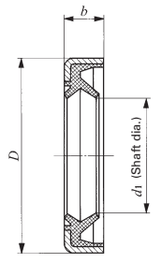


Shaft dia. 20 — 24mm

Shaft dia. mm	Identification number		Boundary dimensions mm			Combinable bearings			
	Single lip	Double lips	d_1	D	b	TA···Z YT TLA···Z YTL	RNA TR TAF GTR	RNAF	NAX NBX
20	**OS 20264**	**DS 20264**	20	26	4	**TLA 2012Z TLA 2016Z TLA 2020Z TLA 2030Z**	—	—	—
	OS 20274	**DS 20274**	20	27	4	**TA 2015Z TA 2020Z TA 2025Z TA 2030Z YT 2015 YT 2025**	—	—	—
	OS 20284	**DS 20284**	20	28	4	**TA 202820Z YT 202820**	**RNA 4902 RNA 6902 TAF 202816 TAF 202820**	**RNAF 202813 RNAFW 202826**	—
	OS 20304	**DS 20304**	20	30	4	—	—	—	**NAX 2030 NBX 2030**
	OS 20324	**DS 20324**	20	32	4	—	—	**RNAF 203212 RNAFW 203224**	—
	OS 20326	**DS 20326**	20	32	6	—	—	**RNAF 203212 RNAFW 203224**	—
21	**OS 21294**	**DS 21294**	21	29	4	**TA 2116Z TA 2120Z YT 2116 YT 2120**	**TAF 212916 TAF 212920**	—	—



OS



DS

Shaft dia. mm	Identification number		Boundary dimensions mm			Combinable bearings			
	Single lip	Double lips	d_1	D	b	TA···Z YT TLA···Z YTL	RNA TR TAF GTR	RNAF	NAX NBX
22	**OS 22284**	**DS 22284**	22	28	4	**TLA 2212Z TLA 2216Z TLA 2220Z**	—	—	—
	OS 22294	**DS 22294**	22	29	4	**TA 2210Z TA 2215Z TA 2220Z TA 2225Z TA 2230Z**	—	—	—
	OS 22304	**DS 22304**	22	30	4	**TA 223016Z TA 223020Z YT 223016 YT 223020**	**RNA 4903 RNA 6903 TAF 223016 TAF 223020**	**RNAF 223013 RNAFW 223026**	—
24	**OS 24314**	**DS 24314**	24	31	4	**TA 2420Z TA 2428Z YT 2428**	—	—	—
	OS 24324	**DS 24324**	24	32	4	**TA 243216Z TA 243220Z YT 243216 YT 243220**	**TAF 243216 TAF 243220**	—	—

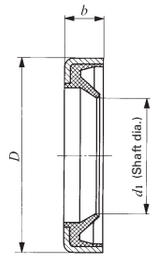
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1mm=0.03937inch

SEALS FOR NEEDLE ROLLER BEARINGS

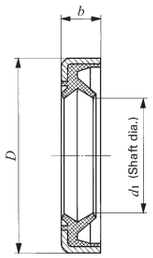


Shaft dia. 25 — 29mm

Shaft dia. mm	Identification number		Boundary dimensions mm			Combinable bearings			
	Single lip	Double lips	d_1	D	b	TA···Z YT TLA···Z YTL	RNA TR TAF GTR	RNAF	NAX NBX
25	**OS 25324**	**DS 25324**	25	32	4	**TLA 2512Z** **TLA 2516Z** **TLA 2520Z** **TLA 2526Z** **TLAW 2538Z** **YTL 2526**	—	—	—
	OS 25334	**DS 25334**	25	33	4	**TA 2510Z** **TA 2515Z** **TA 2520Z** **TA 2525Z** **TA 2530Z** **YT 2510** **YT 2515** **YT 2520** **YT 2525**	**TAF 253316** **TAF 253320**	—	—
	OS 25356	**DS 25356**	25	35	6	—	—	**RNAF 253517** **RNAFW 253526**	—
	OS 25376	**DS 25376**	25	37	6	—	**RNA 4904** **RNA 6904**	**RNAF 253716** **RNAFW 253732**	**NAX 2530** **NBX 2530**
26	**OS 26344**	**DS 26344**	26	34	4	**TA 2616Z** **TA 2620Z** **YT 2616** **YT 2620**	**TAF 263416** **TAF 263420**	—	—



OS



DS

Shaft dia. mm	Identification number		Boundary dimensions mm			Combinable bearings			
	Single lip	Double lips	d_1	D	b	TA···Z YT TLA···Z YTL	RNA TR TAF GTR	RNAF	NAX NBX
28	**OS 28354**	**DS 28354**	28	35	4	**TLA 2816Z** **TLA 2820Z**	—	—	—
	OS 28374	**DS 28374**	28	37	4	**TA 2820Z** **TA 2830Z** **YT 2820**	**TAF 283720** **TAF 283730**	—	—
	OS 28396	**DS 28396**	28	39	6	—	**RNA 49/22** **RNA 69/22**	—	—
	OS 28406	**DS 28406**	28	40	6	—	—	**RNAF 284016** **RNAFW 284032**	—
29	**OS 29384**	**DS 29384**	29	38	4	**TA 2920Z** **TA 2930Z** **YT 2920**	**TAF 293820** **TAF 293830**	—	—

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

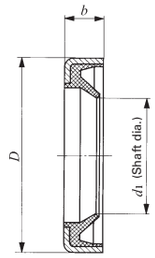
SEALS FOR NEEDLE ROLLER BEARINGS



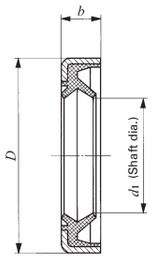


Shaft dia. 30 — 38mm

Shaft dia. mm	Identification number		Boundary dimensions mm			Combinable bearings			
	Single lip	Double lips	d_1	D	b	TA···Z YT TLA···Z YTL	RNA TR TAF GTR	RNAF	NAX NBX
30	OS 30374	DS 30374	30	37	4	TLA 3012Z TLA 3016Z TLA 3018Z TLA 3020Z TLA 3026Z TLAW 3038Z	—	—	—
	OS 30404	DS 30404	30	40	4	TA 3013Z TA 3015Z TA 3020Z TA 3025Z TA 3030Z	TAF 304020 TAF 304030	RNAF 304017 RNAFW 304026	—
	OS 30426	DS 30426	30	42	6	—	RNA 4905 RNA 6905	RNAF 304216 RNAFW 304232	NAX 3030 NBX 3030
32	OS 32424	DS 32424	32	42	4	TA 3220Z TA 3230Z YT 3220	TAF 324220 TAF 324230	—	—
	OS 32456	DS 32456	32	45	6	—	RNA 49/28 RNA 69/28 GTR 324530	—	—



OS



DS

Shaft dia. mm	Identification number		Boundary dimensions mm			Combinable bearings			
	Single lip	Double lips	d_1	D	b	TA···Z YT TLA···Z YTL	RNA TR TAF GTR	RNAF	NAX NBX
35	OS 35424	DS 35424	35	42	4	TLA 3512Z TLA 3516Z TLA 3520Z	—	—	—
	OS 35454	DS 35454	35	45	4	TA 3512Z TA 3515Z TA 3520Z TA 3525Z TA 3530Z	TAF 354520 TAF 354530	RNAF 354517 RNAFW 354526	—
	OS 35476	DS 35476	35	47	6	—	RNA 4906 RNA 6906	RNAF 354716 RNAFW 354732	NAX 3530 NBX 3530
37	OS 37474	DS 37474	37	47	4	TA 3720Z TA 3730Z YT 3720	TAF 374720 TAF 374730	—	—
38	OS 38484	DS 38484	38	48	4	TA 3815Z TA 3820Z TA 3825Z TA 3830Z TAW 3845Z	TAF 384820 TAF 384830	—	—
	OS 38506	DS 38506	38	50	6	—	—	—	—

L

OS
DS

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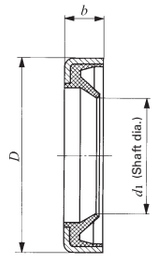
SEALS FOR NEEDLE ROLLER BEARINGS



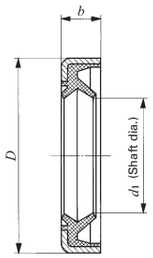


Shaft dia. 40 — 50mm

Shaft dia. mm	Identification number		Boundary dimensions mm			Combinable bearings			
	Single lip	Double lips	d_1	D	b	TA···Z YT TLA···Z YTL	RNA TR TAF GTR	RNAF	NAX NBX
40	**OS 40474**	**DS 40474**	40	47	4	**TLA 4012Z** **TLA 4016Z** **TLA 4020Z**	—	—	—
	OS 40504	**DS 40504**	40	50	4	**TA 4015Z** **TA 4020Z** **TA 4025Z** **TA 4030Z** **TA 4040Z** **YT 4015** **YT 4025**	**TAF 405020** **TAF 405030**	**RNAF 405017** **RNAFW 405034**	—
	OS 40526	**DS 40526**	40	52	6	—	**RNA 49/32** **RNA 69/32**	—	**NAX 4032** **NBX 4032**
	OS 40556	**DS 40556**	40	55	6	—	**TR 405520** **GTR 405520**	**RNAF 405520** **RNAFW 405540**	—
42	**OS 42557**	**DS 42557**	42	55	7	—	**RNA 4907** **RNA 6907**	—	—



OS



DS

Shaft dia. mm	Identification number		Boundary dimensions mm			Combinable bearings			
	Single lip	Double lips	d_1	D	b	TA···Z YT TLA···Z YTL	RNA TR TAF GTR	RNAF	NAX NBX
45	**OS 45524**	**DS 45524**	45	52	4	**TLA 4516Z** **TLA 4520Z**	—	—	—
	OS 45554	**DS 45554**	45	55	4	**TA 4520Z** **TA 4525Z** **TA 4530Z** **TA 4540Z** **YT 4520** **YT 4525**	**TAF 455520** **TAF 455530**	**RNAF 455517** **RNAFW 455534**	—
	OS 45627	**DS 45627**	45	62	7	—	—	**RNAF 456220** **RNAFW 456240**	—
48	**OS 48627**	**DS 48627**	48	62	7	—	**RNA 4908** **RNA 6908** **TR 486230** **GTR 486230**	—	—
50	**OS 50584**	**DS 50584**	50	58	4	**TLA 5020Z** **TLA 5025Z**	—	—	—
	OS 50624	**DS 50624**	50	62	4	**TA 5012Z** **TA 5015Z** **TA 5020Z** **TA 5025Z** **TA 5030Z** **TA 5040Z** **TAW 5045Z**	**TAF 506225** **TAF 506235**	**RNAF 506220** **RNAFW 506240**	**NAX 5035** **NBX 5035**
	OS 50657	**DS 50657**	50	65	7	—	**RNA 49/42**	**RNAF 506520** **RNAFW 506540**	—

L

OS
DS

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

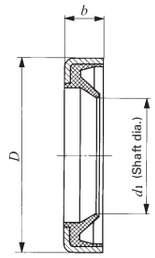
SEALS FOR NEEDLE ROLLER BEARINGS



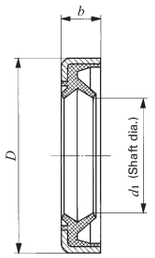


Shaft dia. 52 — 72mm

Shaft dia. mm	Identification number		Boundary dimensions mm			Combinable bearings			
	Single lip	Double lips	d_1	D	b	TA···Z YT TLA···Z YTL	RNA TR TAF GTR	RNAF	NAX NBX
52	OS 52687	DS 52687	52	68	7	—	RNA 4909 RNA 6909	—	—
55	OS 55674	DS 55674	55	67	4	TA 5520Z TA 5525Z TA 5530Z TA 5540Z TAW 5545Z TAW 5550Z	—	—	—
	OS 55687	DS 55687	55	68	7	—	TAF 556825 TAF 556835	RNAF 556820 RNAFW 556840	—
	OS 55727	—	55	72	7	—	—	RNAF 557220 RNAFW 557240	—
58	OS 58727	DS 58727	58	72	7	—	RNA 4910 RNA 6910	—	—
60	OS 60724	DS 60724	60	72	4	TA 6025Z TA 6030Z TA 6040Z TAW 6045Z TAW 6050Z	TAF 607225 TAF 607235	—	NAX 6040 NBX 6040
	OS 60787	DS 60787	60	78	7	—	—	RNAF 607820 RNAFW 607840	—
62	OS 62744	DS 62744	62	74	4	TA 6212Z	—	—	—
	OS 62747	DS 62747	62	74	7	TA 6212Z	—	—	—
63	OS 63807	DS 63807	63	80	7	—	RNA 4911 RNA 6911	—	—



OS



DS

Shaft dia. mm	Identification number		Boundary dimensions mm			Combinable bearings			
	Single lip	Double lips	d_1	D	b	TA···Z YT TLA···Z YTL	RNA TR TAF GTR	RNAF	NAX NBX
65	OS 65774	DS 65774	65	77	4	TA 6525Z TA 6530Z TAW 6545Z TAW 6550Z	—	—	—
	OS 65857	DS 65857	65	85	7	—	—	RNAF 658530 RNAFW 658560	—
68	OS 68857	DS 68857	68	85	7	—	RNA 4912 RNA 6912	—	—
70	OS 70824	DS 70824	70	82	4	TA 7025Z TA 7030Z TA 7040Z TAW 7050Z YT 7025 YT 7030 YT 7040	—	—	—
	OS 70907	DS 70907	70	90	7	—	—	RNAF 709030 RNAFW 709060	—
72	OS 72907	DS 72907	72	90	7	—	RNA 4913 RNA 6913	—	

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

Cir-clips for Needle Roller Bearings

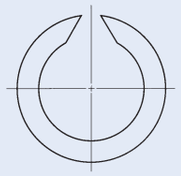
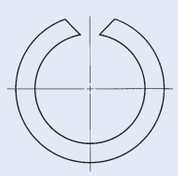
Features

IKO Cir-clips for Needle Roller Bearings have been specially designed for needle roller bearings on which, in many cases, generally available Cir-clips cannot be used. They have a low sectional height and are very rigid. They are made of spring steel.
There are Cir-clips for shafts and for bores, and they are used for positioning to prevent bearing movement in the axial direction.

Types

Cir-clips for Needle Roller Bearings are available as shown in Table. 1.

Table 1 Type of Cir-clip

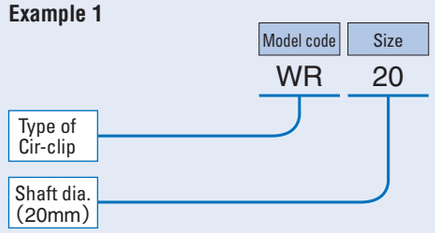
Type	For shaft	For bore
Shape		
Model code	WR	AR

Identification number

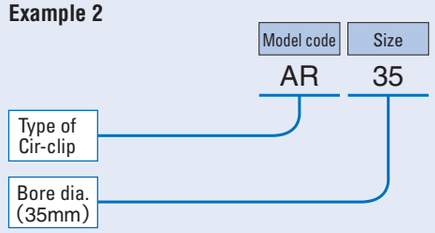
The identification number of Cir-clips consists of a model code and a size as shown below.

Examples of identification number

Example 1



Example 2



Allowable Rotational Speed

Cir-clips for Needle Roller Bearings are fixed in the groove with a certain amount of pressure on the bottom of the groove. In the case of Cir-clips for shaft WR type, the centrifugal force causes a decrease in the gripping pressure. Therefore, when using them at high rotational speeds, it is necessary to first check the allowable rotational speed shown in Fig.1.

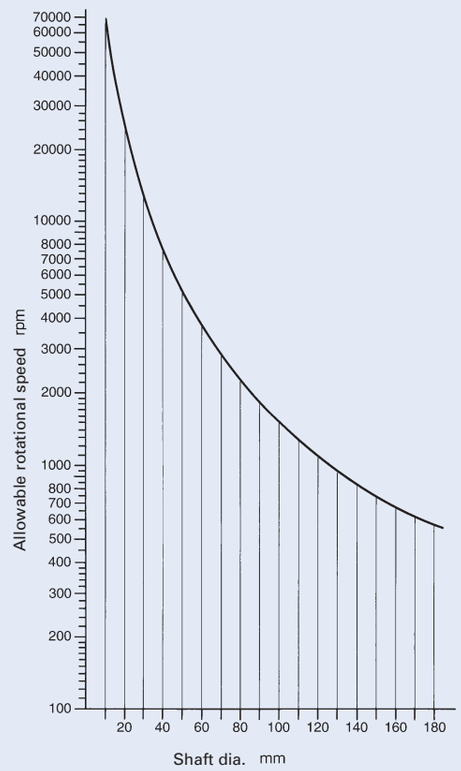


Fig.1 Allowable rotational speed

Mounting

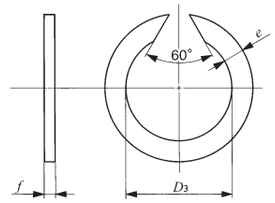
The mounting dimensions for Cir-clips for Needle Roller Bearings are shown in the dimension table.
When using these Cir-clips to restrict the movement of the needle roller cage in the axial direction, it is recommended that a spacer be used between the Cir-clip and the cage. Spacers are not required at low rotational speeds.
When it is difficult to reach Cir-clips with dismounting tools and disassembly is difficult, or when the frequency of dismounting is high, it is necessary to consider the use of a C type retaining ring (JIS B 2804:1978) or C type concentric retaining ring (JIS B 2806:1978), although they have a higher sectional height.

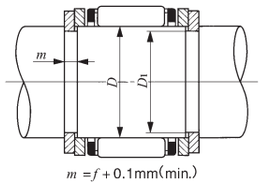
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1mm=0.03937inch

CIR-CLIPS FOR NEEDLE ROLLER BEARINGS

For Shaft







WR

Shaft dia. 4 — 390mm

Identification number	Boundary dimensions mm					
	Shaft dia. D	D_3 (Max.)	e	f	Groove dia. D_1	Tolerance
WR 4	4	3.7	0.8	0.5	3.8	0 — 0.09
WR 5	5	4.7	1	0.5	4.8	
WR 6	6	5.6	1.1	0.7	5.7	
WR 7	7	6.5	1.2	0.7	6.7	
WR 8	8	7.4	1.3	1	7.6	
WR 9	9	8.4	1.3	1	8.6	
WR 10	10	9.4	1.3	1	9.6	
WR 11	11	10.2	1.3	1	10.5	0 — 0.11
WR 12	12	11.2	1.3	1	11.5	
WR 13	13	12.1	1.3	1	12.5	
WR 14	14	13.1	1.5	1.2	13.5	
WR 15	15	14	1.75	1.2	14.4	
WR 16	16	15	1.75	1.2	15.4	
WR 17	17	16	1.75	1.2	16.4	
WR 18	18	17	1.75	1.2	17.4	
WR 19	19	17.9	1.75	1.2	18.4	0 — 0.13
WR 20	20	18.7	1.75	1.2	19.2	
WR 21	21	19.7	1.75	1.2	20.2	
WR 22	22	20.7	1.75	1.2	21.2	
WR 23	23	21.7	1.75	1.2	22.2	
WR 24	24	22.5	1.75	1.2	23	
WR 25	25	23.5	1.75	1.2	24	
WR 26	26	24.5	1.75	1.2	25	
WR 28	28	26.5	2.3	1.5	27	
WR 29	29	27.5	2.3	1.5	28	
WR 30	30	28.5	2.3	1.5	29	
WR 32	32	30.2	2.3	1.5	30.8	0 — 0.16
WR 35	35	33.2	2.3	1.5	33.8	
WR 36	36	34.2	2.3	1.5	34.8	
WR 37	37	35.2	2.3	1.5	35.8	
WR 38	38	36.2	2.3	1.5	36.8	
WR 40	40	37.8	2.3	1.5	38.5	

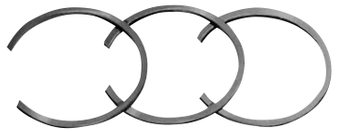
Identification number	Boundary dimensions mm					
	Shaft dia. D	D_3 (Max.)	e	f	Groove dia. D_1	Tolerance
WR 42	42	39.8	2.3	1.5	40.5	0 — 0.16
WR 43	43	40.8	2.3	1.5	41.5	
WR 45	45	42.8	2.3	1.5	43.5	
WR 47	47	44.8	2.3	1.5	45.5	
WR 50	50	47.8	2.3	1.5	48.5	
WR 52	52	49.8	2.3	1.5	50.5	0 — 0.19
WR 55	55	52.6	2.3	1.5	53.5	
WR 60	60	57.6	2.3	1.5	58.5	
WR 63	63	60.6	2.3	1.5	61.5	
WR 65	65	62.6	2.3	1.5	63.5	
WR 68	68	65.4	2.8	2	66.2	
WR 70	70	67.4	2.8	2	68.2	
WR 75	75	72.4	2.8	2	73.2	
WR 80	80	77.4	2.8	2	78.2	
WR 82	82	79.3	3.4	2.5	80.2	0 — 0.22
WR 85	85	82	3.4	2.5	83	
WR 90	90	87	3.4	2.5	88	
WR 95	95	92	3.4	2.5	93	
WR 100	100	97	3.4	2.5	98	
WR 105	105	101.7	3.4	2.5	102.7	
WR 110	110	106.7	3.4	2.5	107.7	
WR 115	115	111.7	3.4	2.5	112.7	
WR 120	120	116.7	3.4	2.5	117.7	
WR 125	125	121.7	3.4	2.5	122.7	0 — 0.25
WR 130	130	126.7	3.4	2.5	127.7	
WR 135	135	131.6	4	2.5	132.4	
WR 140	140	136.6	4	2.5	137.4	
WR 145	145	141.6	4	2.5	142.4	
WR 150	150	146.6	4	2.5	147.4	
WR 155	155	151.6	4	2.5	152.4	
WR 160	160	156.6	4	2.5	157.4	
WR 165	165	161.6	4	2.5	162.4	

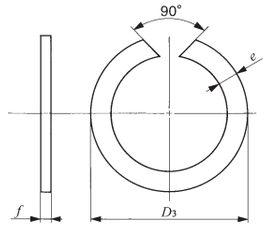
Identification number	Boundary dimensions mm					
	Shaft dia. D	D_3 (Max.)	e	f	Groove dia. D_1	Tolerance
WR 170	170	166.6	4	2.5	167.4	0 — 0.25
WR 175	175	171.6	4	2.5	172.4	
WR 180	180	175.6	5	3	177	
WR 185	185	180.6	5	3	182	
WR 190	190	185.6	5	3	187	
WR 195	195	190.6	5	3	192	
WR 200	200	195.6	5	3	197	0 — 0.29
WR 210	210	205.6	5	3	207	
WR 220	220	215.6	5	3	217	
WR 230	230	225.6	5	3	227	
WR 240	240	235.6	5	3	237	
WR 260	260	253	7.5	4	255	0 — 0.32
WR 265	265	258	7.5	4	260	
WR 270	270	263	7.5	4	265	
WR 280	280	273	7.5	4	275	
WR 285	285	278	7.5	4	280	
WR 300	300	293	7.5	4	295	
WR 305	305	298	7.5	4	300	
WR 320	320	313	7.5	4	315	
WR 330	330	323	7.5	4	325	0 — 0.36
WR 340	340	333	7.5	4	335	
WR 350	350	343	7.5	4	345	
WR 360	360	353	7.5	4	355	
WR 370	370	363	7.5	4	365	
WR 390	390	383	7.5	4	385	

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

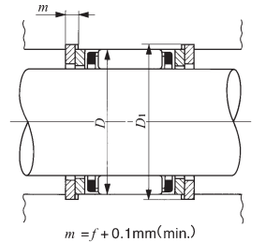
CIR-CLIPS FOR NEEDLE ROLLER BEARINGS

For Bore





AR



$m = f + 0.1$mm(min.)

Bore dia. 7 — 440mm

Identification number	Boundary dimensions mm					
	Bore dia. D	D_3 (Min.)	e	f	Groove dia. D_1	Tolerance
AR 7	7	7.5	1	0.8	7.3	+0.09 0
AR 8	8	8.5	1	0.8	8.3	
AR 9	9	9.5	1.1	0.8	9.3	
AR 10	10	10.6	1.2	0.8	10.4	
AR 11	11	11.6	1.3	1	11.4	
AR 12	12	12.7	1.3	1	12.4	
AR 13	13	13.8	1.3	1	13.5	+0.11 0
AR 14	14	14.8	1.3	1	14.5	
AR 15	15	15.8	1.3	1	15.5	
AR 16	16	16.8	1.6	1.2	16.5	
AR 17	17	17.8	1.6	1.2	17.5	
AR 18	18	18.9	1.75	1.2	18.5	
AR 19	19	19.9	1.75	1.2	19.6	
AR 20	20	21	1.75	1.2	20.6	
AR 21	21	22	1.75	1.2	21.6	
AR 22	22	23	1.75	1.2	22.6	
AR 23	23	24	1.75	1.2	23.6	+0.13 0
AR 24	24	25.2	1.75	1.2	24.8	
AR 25	25	26.2	1.75	1.2	25.8	
AR 26	26	27.2	1.75	1.2	26.8	
AR 27	27	28.2	1.75	1.2	27.8	
AR 28	28	29.2	1.75	1.2	28.8	
AR 29	29	30.2	1.75	1.2	29.8	
AR 30	30	31.4	2.3	1.5	31	
AR 31	31	32.4	2.3	1.5	32	
AR 32	32	33.4	2.3	1.5	33	
AR 33	33	34.4	2.3	1.5	34	
AR 34	34	35.4	2.3	1.5	35	+0.16 0
AR 35	35	36.4	2.3	1.5	36	
AR 37	37	38.8	2.3	1.5	38.2	
AR 38	38	39.8	2.3	1.5	39.2	
AR 39	39	40.8	2.3	1.5	40.2	

Identification number	Boundary dimensions mm					
	Bore dia. D	D_3 (Min.)	e	f	Groove dia. D_1	Tolerance
AR 40	40	41.8	2.3	1.5	41.2	+0.16 0
AR 42	42	43.8	2.3	1.5	43.2	
AR 43	43	44.8	2.3	1.5	44.2	
AR 44	44	45.8	2.3	1.5	45.2	
AR 45	45	46.8	2.3	1.5	46.2	
AR 47	47	48.8	2.3	1.5	48.2	
AR 48	48	49.8	2.3	1.5	49.2	
AR 50	50	51.8	2.3	1.5	51.2	
AR 52	52	54.3	2.3	1.5	53.5	
AR 53	53	55.3	2.3	1.5	54.5	
AR 55	55	57.3	2.3	1.5	56.5	
AR 57	57	59.3	2.3	1.5	58.5	
AR 58	58	60.3	2.3	1.5	59.5	
AR 60	60	62.3	2.3	1.5	61.5	
AR 62	62	64.3	2.3	1.5	63.5	+0.19 0
AR 65	65	67.3	2.3	1.5	66.5	
AR 68	68	70.3	2.3	1.5	69.5	
AR 70	70	72.3	2.3	1.5	71.5	
AR 72	72	74.6	2.8	2	73.8	
AR 73	73	75.6	2.8	2	74.8	
AR 75	75	77.6	2.8	2	76.8	
AR 76	76	78.6	2.8	2	77.8	
AR 78	78	80.6	2.8	2	79.8	
AR 80	80	82.6	2.8	2	81.8	
AR 81	81	83.6	2.8	2	82.8	
AR 82	82	84.6	2.8	2	83.8	
AR 83	83	85.6	2.8	2	84.8	+0.22 0
AR 85	85	87.6	2.8	2	86.8	
AR 86	86	88.6	2.8	2	87.8	
AR 88	88	91	3.4	2.5	90	
AR 90	90	93	3.4	2.5	92	
AR 92	92	95	3.4	2.5	94	

Identification number	Boundary dimensions mm					
	Bore dia. D	D_3 (Min.)	e	f	Groove dia. D_1	Tolerance
AR 93	93	96	3.4	2.5	95	
AR 95	95	98	3.4	2.5	97	
AR 97	97	100	3.4	2.5	99	
AR 98	98	101	3.4	2.5	100	
AR 100	100	103	3.4	2.5	102	
AR 102	102	105.3	3.4	2.5	104.3	
AR 103	103	106.3	3.4	2.5	105.3	+0.22 0
AR 105	105	108.3	3.4	2.5	107.3	
AR 107	107	110.3	3.4	2.5	109.3	
AR 108	108	111.3	3.4	2.5	110.3	
AR 110	110	113.3	3.4	2.5	112.3	
AR 112	112	115.3	3.4	2.5	114.3	
AR 113	113	116.3	3.4	2.5	115.3	
AR 115	115	118.3	3.4	2.5	117.3	
AR 117	117	120.3	3.4	2.5	119.3	
AR 118	118	121.3	3.4	2.5	120.3	
AR 120	120	123.3	3.4	2.5	122.3	
AR 123	123	126.3	3.4	2.5	125.3	
AR 125	125	128.3	3.4	2.5	127.3	
AR 127	127	130.3	3.4	2.5	129.3	
AR 130	130	133.3	3.4	2.5	132.3	
AR 133	133	136.3	3.4	2.5	135.3	
AR 135	135	138.3	3.4	2.5	137.3	+0.25 0
AR 137	137	140.3	3.4	2.5	139.3	
AR 140	140	143.6	4	2.5	142.6	
AR 143	143	146.6	4	2.5	145.6	
AR 145	145	148.6	4	2.5	147.6	
AR 150	150	153.6	4	2.5	152.6	
AR 153	153	156.6	4	2.5	155.6	
AR 160	160	163.6	4	2.5	162.6	
AR 163	163	166.6	4	2.5	165.6	
AR 165	165	168.6	4	2.5	167.6	

Identification number	Boundary dimensions mm					
	Bore dia. D	D_3 (Min.)	e	f	Groove dia. D_1	Tolerance
AR 170	170	173.6	4	2.5	172.6	+0.25 0
AR 173	173	176.6	4	2.5	175.6	
AR 175	175	178.6	4	2.5	177.6	
AR 180	180	183.6	4	2.5	182.6	
AR 183	183	186.6	4	2.5	185.6	
AR 190	190	194.5	5	3	193	
AR 195	195	199.5	5	3	198	
AR 200	200	204.5	5	3	203	
AR 205	205	209.5	5	3	208	+0.29 0
AR 210	210	214.5	5	3	213	
AR 215	215	219.5	5	3	218	
AR 220	220	224.5	5	3	223	
AR 225	225	229.5	5	3	228	
AR 230	230	234.5	5	3	233	
AR 235	235	239.5	5	3	238	
AR 240	240	244.5	5	3	243	+0.32 0
AR 245	245	249.5	5	3	248	
AR 250	250	254.5	5	3	253	
AR 260	260	267	7.5	4	265	
AR 270	270	277	7.5	4	275	
AR 280	280	287	7.5	4	285	
AR 300	300	307	7.5	4	305	
AR 320	320	327	7.5	4	325	+0.36 0
AR 325	325	332	7.5	4	330	
AR 355	355	362	7.5	4	360	
AR 375	375	382	7.5	4	380	
AR 395	395	402	7.5	4	400	
AR 415	415	422	7.5	4	420	+0.4 0
AR 420	420	427	7.5	4	425	
AR 440	440	447	7.5	4	445	

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch